Costmap vs Height Map vs Occupancy Grid (ROS 2)

Aspect	**Costmap**	**Height Map (Elevation)**	**Occupancy Grid (OG)**
What it stores	Traversal *cost* per cell (ease/difficulty)	Ground *height* in meters per cell (2.5D)	*Probability* that a cell is occupied
Typical ROS msg	Often published as `nav_msgs/OccupancyGrid` (int8 costs)	`sensor_msgs/Image` (e.g., 32FC1) for viz, or `grid_map_msgs/GridMap` (*recommended*)	`nav_msgs/OccupancyGrid` (standard OG)
Value range / units	Int in [0..100] (unitless cost; 100 often lethal)	Float (meters), e.g., 0.0–4.0 m	Int in {-1,0..100}: -1 unknown, 0 free, 100 occupied
Geometry metadata	Yes (resolution, width, height, origin)	Yes with GridMap; plain `Image` has no geometry	Yes (resolution, width, height, origin)
How RViz renders	Map display: *Costmap* or *Raw*	Image display (`32FC1`); GridMap plugin colorizes layers	Map display: *Map/Trinary/Raw*
Typical sources	Inflated obstacles, slope/roughness, keepouts	Stereo/LiDAR DEMs, depth fusion, terrain mapping	SLAM/static map server, raytracing sensors
Used by	Local/global planners (minimize path cost)	Terrain analysis; derive traversability/cost layers	Localization, global planning, static world model
Common pitfalls	Confusing cost with meters; accidental 100 (lethal)	Publishing as plain Image (no geometry); frame/origin mismatch	Forgetting -1 (unknown); treating OG like a costmap

Typical mappings from a grayscale PNG (pixel $g \in [0, 255]$)

$$\textbf{Height (meters):}\quad h = \frac{g}{255}\, H_{\max} \quad \text{(publish as 32FC1 or GridMap elevation)}$$

$$\textbf{Cost (unitless):}\quad c = \mathrm{clamp}\left(\left\lfloor \frac{g}{255}\, C_{\max} + 0.5 \right\rfloor,\ 0,\ C_{\max} \right), \quad C_{\max} \in \{98, 99, 100\}$$

$$\textbf{Occupancy (OG):}\quad \mathrm{occ} = \begin{cases} 100 & g < \tau_{\mathrm{occ}} \quad \text{(occupied)} \\ 0 & g \geq \tau_{\mathrm{free}} \quad \text{(free)} \\ -1 & \text{otherwise (unknown)} \end{cases}$$

How your three topics fit

- **/costmap** (`nav_msgs/OccupancyGrid`): *cost* per cell (ints). Used by planners.
- **/height_image** (`sensor_msgs/Image`, 32FC1): *height in meters* per pixel (smooth viz).
  Tip: Use GridMap for geometry-aware elevation layers that overlay in the map frame.
- **/costmap_image** (`sensor_msgs/Image`, mono8): the original grayscale for exact visual fidelity.

RViz tips

- Costmap: Map display → Color Scheme=*Costmap* (or *Raw* to see 0–100 grayscale).
- Height image: Image display on `/height_image` (32FC1), enable *Normalize Range*.
- For elevation that overlays with geometry, prefer GridMap over plain Image.

Gotchas

- Capping cost to 4 *does not* make a height map; it just yields 5 cost bands (0–4).

- Publishing cost 100 unintentionally can mark cells as lethal/blocked.
- Convert OpenCV (top-left) to ROS grid (bottom-left) by flipping rows for OccupancyGrid.